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are valued, softer skills of concept selling and relationship management are also required by the bigger OEM companies. Vivek Madhukar, COO, TBSL (which operates TimesJobs.com) explained, "Automotive electronics is a highly specialised field, and companies seek very specific profiles in this domain—offering high starting salaries and good career growth."

According to the data provided by TimesJobs.com, the top industries hiring verification and validation engineers form the core automobile electronics industry with about 73 per cent share of jobs, followed by the IT software/software services industry with a share of 13 per cent. IT-hardware/semiconductor and petrochemicals/oil and gas/power industries have 8 per cent and 5 per cent of the jobs, respectively.

Apart from Delhi and Chennai that feature in the list of top locations for jobs in automotive electronics field with 16 per cent and 12 per cent, respectively, Bangalore, Mumbai and Hyderabad compete with 11 per cent, 10 per cent and 10 per cent of the jobs, respectively. Remaining jobs are spread in other locations.

Pay package

According to the stats provided by TimesJobs.com, 49 per cent jobs are for freshers with a pay package between ₹ 190,000 and ₹ 450,000 per annum. Twenty two per cent of the jobs are for professionals with at least two to three years of experience, who are paid between ₹ 500,000 and ₹ 1,000,000. Eleven per cent of the jobs available in India are for mid-level experienced professionals whereas the remaining 18 per

cent of the jobs are for seniors in this industry.

As found on payscale.com, the average pay for an automotive engineer is ₹ 390,652 per annum. RBEI recruits mainly from Tier-1 and Tier-2 engineering colleges, and these cater to most of the software, hardware and mechanical design activities. The IITs and NITs cater to around 5 per cent of the annual requirement, and these engineers engage in tasks demanding high ability to solve core engineering problems by delving deep into engineering principles and designs systems through creative and innovative thinking.

Therefore, Garadi says, "Pay packages vary too much in the industry for any price point to be relevant. However, we can say that it is comparable to any other engineering domain."

Skills expected

Modern-day industries, including automotive, demand multi-skilled engineers handy with both core engineering as well as IT engineering. This has led to the birth of a new branch of science, particularly popular in the automotive industry, called mechatronics. It is a combination of mechanical, electronics, control and computer engineering. Almost every subsystem in a vehicle is moving from a pure mechanical system to a mechatronic system. Garadi says, "Therefore future of the automotive industry lies in the confluence of multiple fields of engineering, and an engineer aspiring to score in the automotive industry has to be multi-skilled or, in other words, should be a mechatronics engineer."

Ravindra adds that, "For a fresher, we expect the person to be strong in communication standards, C/C++, microcontrollers and MEMS-based sensor technologies. Experienced graduates in this domain will need experience in developing hardware/software as per established standards of the automotive industry; hands-on experience in one or more programming languages and experience in building real-time operating system (RTOS) based products amongst others."

Expert advice and suggestions

These days a big chunk of engineers recruited in the automotive domain broadly fit into the IT category. They write software code to solve real-world problems and implement systems. These software can run on desktop computers or on on-board computers (also referred to as ECUs). They deal with real-world data and signals, and interpret, analyse and process these to realise complex functions. Garadi explains that, "For these functions to be effective, efficient, accurate, repeatable and reliable, engineers have to apply mathematical, scientific and theoretical skills." They need sound theoretical approach for introduction of new ideas and concepts.

All of these amount to core engineering skills. IT engineers, in order to make a good job of what they are working on, should be in a position to understand, visualise and appreciate the engineering principles. More complex the systems are, deeper you may have to delve. Lest, the work-piece would only turn out mediocre. Garadi says, "For example, the engineers engaged in developing software for the vehicle exhaust management need to understand the principles behind chemical processes, fluid dynamics, thermal engineering, etc. Likewise, engineers writing software for the engine management systems would do well to have basic engineering skills associated with combustion, kinetics, kinematics, traction, etc."

On another note, Ravindra says, "For people interested in the field of automotive electronics, there is currently a lot of excitement with several new technologies being developed and integrated into future automotive devices. It is important that the graduates be strong in fundamentals of communication software engineering since the product developed will ultimately go into a vehicle, and this involves the safety of the people on board." ●

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